SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

Question:20

Duration:30min

Total marks:20

Annexure A MCQ

SCENARIO BASED

Code of Conduct:

I B.Deepthi-(192424269) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B.Deepthi
Signature of the student:

Scenario Based Assignment – Artificial Intelligence

1. AI-Powered Drone for Emergency Medicine Delivery

i. Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the path that appears closest to the destination based only on the heuristic value (h(n)), such as straight-line distance. It ignores the actual travel cost.

Strengths:

- Fast decision-making.
- Requires less computation.
- Suitable for emergency situations where quick responses are required.

Weaknesses:

- Does not guarantee the shortest or safest path.
- May choose blocked or risky routes.
- Can get trapped in dead ends because it ignores actual path cost.

*ii. A Search**

A* Search uses the evaluation function:

f(n) = g(n) + h(n)

Where:

- **g(n)** = Actual cost from the start.
- **h(n)** = Estimated cost to the goal.

A* balances both the travel cost and estimated distance, helping the drone choose safer and more efficient routes.

Advantages:

- Finds the optimal path when the heuristic is admissible.
- Avoids unnecessarily expensive routes.
- Adapts better to changing conditions.

iii. Impact of Heuristic Accuracy

- **Accurate heuristic:** Faster search and better route selection.
- **Overestimated heuristic:** A* may lose optimality.

  Poor heuristic: Both algorithms explore more unnecessary nodes.

Greedy Best-First Search depends entirely on heuristic accuracy, whereas A* is less affected because it also considers actual cost.

iv.Effect of Dynamic Changes

Blocked roads and changing weather require replanning.

Greedy Best-First Search

- Frequently chooses blocked routes.
- Needs repeated searching.

A*

- Recalculates costs efficiently.
- Produces better alternative routes.
- More reliable under changing conditions.

v.Recommended

Algorithm

Recommended: *A*

*Search** **Justification:**

- Produces optimal routes.
- Considers safety and travel cost.
- Performs better in dynamic environments.
- Provides reliable decisions for emergency medicine delivery.

2. Smart City Traffic Signal

Optimization Problem Formulation

This is an optimization problem.

Objective:

- Minimize waiting time.
- Minimize fuel consumption.
- Reduce traffic congestion.

Traditional search algorithms are inefficient because:

- The search space is extremely large.
- Traffic changes continuously.
- Exhaustive search requires excessive computation.

i. Hill Climbing

Hill Climbing starts with an initial traffic signal configuration and continuously selects a neighboring solution with better performance.

Advantages

- Simple implementation.
- Fast execution.
- Low memory usage.

Limitations

- Can become trapped in local optimum.
- Cannot guarantee the best traffic schedule.

ii. Local Maxima, Plateaus and

Ridges Local Maximum

- AI believes traffic flow is optimal although a better solution exists.

Plateau

- Multiple signal timings produce nearly identical performance.
- No clear improvement direction.

Ridge

- Improvement requires several coordinated signal changes.
- Hill Climbing struggles to move toward the better solution.

iii.Simulated Annealing and Genetic Algorithms Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.
- Finds better global solutions.

Genetic Algorithm

Uses population-based search.

- Applies selection, crossover and mutation.
- Explores many solutions simultaneously.
- Highly suitable for large optimization problems.

iv.Recommended Approach

Recommended: Genetic Algorithm Justification

- Highly scalable.
- Handles changing traffic efficiently.
- Avoids local optima.
- Produces near-optimal signal timings.

3. University Exam Timetable (Constraint Satisfaction Problem) Problem Formulation

Variables

Exam subjects

- Time slots
- Exam halls

Domains

Possible values for:

- Available time slots
- Available rooms

Constraints

- No overlapping exams.
- Limited hall capacity.
- Faculty availability.
- Subject precedence.

i. Variables, Domains and Constraints

Variables represent exams.

Domains represent possible time slots and halls.

Constraints ensure valid scheduling

ii. Backtracking Search

Steps:

1. Assign a subject to a time slot.
2. Check constraints.
3. If constraints fail, undo assignment.
4. Try another assignment.
5. Continue until all exams are scheduled.

iii. Forward Checking

Forward Checking:

- Removes invalid choices early.
- detects conflicts before deeper search.
- Reduces unnecessary computation.
- Speeds up timetable generation.

iv.Real-world Challenges

Challenges:

- Thousands of students.
- Hundreds of subjects.
- Faculty scheduling.
- Hall shortages.
- Continuous timetable modifications.

Best Strategy

Backtracking + Forward Checking + Constraint Propagation This combination improves scalability and efficiency.

v.Recommended Strategy

Recommended: Minimax + Alpha-Beta Pruning + Evaluation Function

Justification

- Produces near-optimal decisions.
- Reduces computation.
- Meets real-time constraints.
- Performs efficiently in complex strategy games.

Conclusion

Artificial Intelligence algorithms solve different real-world problems based on their strengths. A* Search is ideal for dynamic pathfinding, Genetic Algorithms excel in optimization, Online Search Agents handle uncertain environments, CSP efficiently solves scheduling problems, and Minimax with Alpha-Beta Pruning enables intelligent real-time game decision-making.